

Nirma University

Institute of Technology

Semester End / Supplementary Examination (IR/SPE), February - 2022

Master of Computer Application, Semester - I

3MCA104 Database Management System

Roll /
Exam No.

Supervisor's initial
with date

Time: 2 Hours

Max. Marks: 50

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.

Q-1. Answer the following. [18]

- (A) Discuss union, intersect, and except operations over two sets by taking the appropriate examples of SQL. (6)
CO-2, BL-4
- (B) Discuss simplistic lock, pre-claiming lock, two-phase locking, and strict two-phase locking. (6)
CO-3, BL-1
- (C) Why do we need a SQL key in the database management system? Discuss super key, candidate key, composite key, and alternative key by taking an appropriate example. (6)
CO-2, BL-1

OR

- (C) What do you understand by data model? Discuss the different categories of data models. (6)
CO-1, BL-2

Q-2. Answer the following. [14]

- (A) Discuss the various components of a database system and the connections among them with an appropriate diagram. (7)
CO-1, BL-4
- (B) Why normalization process in a database management system is important? (7)
CO-2, BL-4
- Discuss first normal form(1NF), second normal form(2NF), and third normal form(3NF) with appropriate examples.

Q-3. Answer the following. [18]

- (A) What is the objective of creating a trigger? Write a PL/SQL program to demonstrate a trigger for inserting data in a table. Assume and mention the necessary data if required. (9)
CO-4, BL-6

- (B) Write SQL for creating a table corresponding to the schema as mentioned below: (9)
CO-3, BL-6 department(dept name, building, budget)
course(course id, title, dept name, credits)
instructor(ID, name, dept name, salary)
section(course id, sec id, semester, year, building, room number, time slot id)
teaches(ID, course id, sec id, semester, year)

Make any reasonable assumptions about data types, primary key and reference key.

OR

- (B) What do you understand by context area? Briefly discuss the attributes of (9)
CO-4, BL-2 the implicit cursor. Also, Write a PL/SQL program that update the table and increase the salary of each customer by 500 and use the SQL%ROWCOUNT attribute to determine the number of rows affected. Assume and mention the necessary data if required.
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